

Energy-efficient UAV Swarm Multi-Target Tracking Trajectory Optimization

팀: 날파리
김지환, 신동주, 변혁준, 이지훈
지도교수: 이수경
담당 대학원생: 김창경

Contents

1	Introduction	2
1.1	Background and Motivation	2
1.1.1	Unmanned Aerial Vehicles (UAVs)	2
1.1.2	Integrated Sensing and Communication (ISAC)	2
1.1.3	Agentic AI (AAI)	2
1.2	Limitations of Existing Work	2
2	System Model	3
2.1	System Architecture	3
2.2	Energy Model	3
2.2.1	Sensing Energy	4
2.2.2	Communication Energy	4
2.2.3	Propulsion Energy	4
2.3	Target State Model, Extended Kalman Filter (EKF), and State Fusion	5
2.4	Tracking Accuracy Metric	5
2.5	Problem Formulation	5
3	Proposed Approach	5
4	Preliminary Results	6
5	Current Progress	6
6	Future Plans	7
	References	8

1 Introduction

1.1 Background and Motivation

1.1.1 Unmanned Aerial Vehicles (UAVs)

UAVs have emerged as an effective platform for Multi-Target Tracking (MTT) missions, overcoming limited coverage and signal blockage issues that constrain conventional terrestrial tracking systems [1, 2]. Their 3D mobility and deployment flexibility allow UAVs to establish strong Line-of-Sight (LoS) links and extend coverage over complex terrains [1]. Industry platforms such as Palladyne AI's Pilot and SwarmOS have already demonstrated autonomous multi-drone target tracking with adaptive reassignment in real-world deployments. However, operating MTT on UAV platforms introduces unique challenges, including rapid angular changes, target occlusion in cluttered environments, and the small apparent size of targets observed from high altitude [1]. Most critically, UAVs are severely constrained by limited onboard batteries that must simultaneously power flight propulsion, radar sensing, and data communication. As a result, poor trajectory planning can lead to premature energy depletion, shortened mission duration, and degraded tracking accuracy, motivating the need for energy-efficient trajectory optimization frameworks.

1.1.2 Integrated Sensing and Communication (ISAC)

ISAC has emerged as a key technology for 6G wireless networks, where infrastructure and spectrum resources are shared by both sensing and communication services. By unifying both functionalities into a single transceiver, ISAC provides high throughput, low latency, and highly reliable communication along with accurate sensing [3]. This dual functionality has shown great potential when deployed on aerial platforms, especially UAVs. The dynamic nature of UAVs offers a solution for maintaining strong LoS links, effectively mitigating signal blockage caused by urban structures [2]. By leveraging UAV mobility, ISAC-enabled UAVs can serve as efficient surveillance platforms and target tracking agents capable of following uncooperative targets [1, 2, 3, 4, 5].

1.1.3 Agentic AI (AAI)

AAI refers to an autonomous, goal-driven system that operates with minimal human intervention through continuous perception-reasoning-action loops [6, 7]. Unlike traditional AI, AAI exhibits high adaptability and self-sufficiency. Enabled by a Large Language Model (LLM), it achieves contextual comprehension and structured reasoning, allowing autonomous goal-directed behavior [7]. As 6G networks demand ultra-low latency and efficient resource management, AAI addresses these challenges by adjusting strategies based on real-time feedback and leveraging accumulated experience to refine its decisions [7]. This makes AAI well suited for resource-constrained aerial networks, where agents must coordinate sensing, communication, and mobility simultaneously under rapidly changing conditions [6].

1.2 Limitations of Existing Work

Existing research on UAV-based target tracking has made significant progress in trajectory optimization and radar resource allocation, but several limitations remain.

- **Single-UAV limitation:** Zhao *et al.* consider only a single UAV for target tracking, which restricts the system's ability to maintain continuous coverage over multiple targets [4].
- **Decoupled sensing and communication:** Yang *et al.* treats sensing and communication as separate modules, failing to leverage the joint resource optimization that ISAC provides, which results in redundant spectrum usage and suboptimal energy efficiency [1].

- **Lack of a unified energy model:** Zhou *et al.* leverages ISAC in UAV swarms for cooperative MTT, yet the framework fails to incorporate a unified energy model that accounts for propulsion cost, risking premature mission termination in energy-constrained operation [5].
- **No joint optimization under AAI:** Nguyen *et al.* demonstrates the potential of AAI for autonomous decision-making in UAV swarms through LLM-driven perception-reasoning-action loops, but does not address joint optimization of sensing, communication, and propulsion energy under tracking constraints [6].

In summary, existing works address individual aspects of these challenges but lack a unified framework that jointly considers multi-UAV coordination, ISAC resource sharing, and comprehensive energy optimization while adapting to dynamically changing conditions. Moreover, real-world MTT missions involve challenges such as UAV energy depletion, communication link degradation, and targets approaching sensitive areas, which static optimization frameworks cannot handle.

To address these limitations, this research designs an energy-efficient trajectory optimization framework for a swarm of ISAC-enabled UAVs performing MTT, integrating an AAI module capable of real-time contextual reasoning and dynamic constraint adjustment.

2 System Model

2.1 System Architecture

We consider a multi-UAV assisted MTT system coordinated by a Base Station (BS) located at the origin. The system comprises a set $\mathcal{U} = \{1, 2, \dots, U\}$ of UAVs and a set $\mathcal{K} = \{1, 2, \dots, K\}$ of mobile ground targets. The operational area also contains a set $\mathcal{Z} = \{1, 2, \dots, Z\}$ of critical zones that represent sensitive civilian areas requiring heightened surveillance.

The BS operates an AAI framework that monitors evolving mission conditions and dynamically reassigns targets to UAVs. Each UAV then independently computes its next waypoint using an onboard Multi-Agent Deep Reinforcement Learning (MADRL) policy, trained offline at the BS under Centralized Training and Decentralized Execution (CTDE) paradigm. The time horizon T is divided into equal time slots of duration δ , each split into two sub-slots for sensing and communication, respectively. Each UAV flies at fixed altitude H with maximum speed $v_{\max}$ and tracks at most one target per time slot. The position of UAV u at time t is $\mathbf{q}_{u,t} = [x_{u,t}, y_{u,t}]^\top$, the position of target k at time t is $\mathbf{q}_{k,t} = [x_{k,t}, y_{k,t}]^\top$ and the positions of the critical zones $\{\mathbf{q}_z\}_{z \in \mathcal{Z}}$ are assumed to be known a priori at the BS. In each time slot, every UAV senses its assigned target to estimate the target's position and velocity, then transmits the local estimate along with its own position and remaining energy to the BS. The BS fuses all local estimates into a global estimate for each target. Based on the fused estimates, UAV states, tracking accuracy, and each target's proximity to critical zones, the AAI framework at the BS leverages a LLM to perform contextual reasoning and dynamically reassign targets to UAVs as mission conditions evolve. Once the BS broadcasts the new assignments and global estimate, each UAV independently computes its next waypoint using its onboard MADRL policy and moves to the new position, completing the time slot.

2.2 Energy Model

The total energy consumption of each UAV per time slot consists of three components: sensing, communication, and propulsion. We denote the total energy of UAV u at time slot t as

$$E_{u,t}^{\text{tot}} = E_{u,t}^{\text{sen}} + E_{u,t}^{\text{comm}} + E_{u,t}^{\text{move}}. \quad (1)$$

2.2.1 Sensing Energy

During the first sub-slot, each UAV emits a radar signal with transmit power p for a dwell time τ_s to detect its assigned target. The Signal-to-Noise Ratio (SNR) of the echo reflected from target k is given by [4], [8], [9]:

$$\gamma_{u,k,t}^{\text{sen}} = \frac{pG_t G_r \lambda^2 \sigma \tau_s}{N_0 (4\pi)^3 d_{u,k,t}^4}, \quad (2)$$

where G_t and G_r are the transmitter and receiver antenna gains, $\lambda = c/f_c$ is the wavelength, σ is the target radar cross section, N_0 is the noise power, and $d_{u,k,t} = \sqrt{\|\mathbf{q}_{u,t} - \mathbf{q}_{k,t}\|^2 + H^2}$ is the UAV to target distance. The corresponding sensing energy consumption is [10]

$$E_{u,t}^{\text{sen}} = p\tau_s. \quad (3)$$

2.2.2 Communication Energy

During the second sub-slot, each UAV transmits its local target estimate and status information to the BS. The channel between UAV u and the BS follows a probabilistic LoS/Non-Line-of-Sight (NLoS) model, where the LoS probability is given by [11]

$$P_{u,t}^{\text{LoS}} = (1 + c_1 \exp[-c_2 (\theta_{u,t} - c_1)])^{-1}, \quad (4)$$

where $\theta_{u,t} = \arctan\left(\frac{H}{d_{u,t}}\right)$ is the elevation angle, $d_{u,t} = \|\mathbf{q}_{u,t} - \mathbf{q}_b\|$ is the horizontal distance from the UAV to the BS, and c_1, c_2 are environment dependent constants. The average channel power gain is then [11]

$$h_{u,t}^c = \frac{(P_{u,t}^{\text{LoS}} \eta^{\text{LoS}} + P_{u,t}^{\text{NLoS}} \eta^{\text{NLoS}}) \lambda^2}{(4\pi)^2 d_{u,t}^2}, \quad (5)$$

where $P_{u,t}^{\text{NLoS}} = 1 - P_{u,t}^{\text{LoS}}$ and η^{LoS} and η^{NLoS} are additional attenuation factors for LoS and NLoS links. The communication rate is [4]

$$R_{u,t}^c = \alpha_{u,k,t} \tau_c B_b \log_2(1 + \gamma_{u,t}^c), \quad (6)$$

where τ_c is the communication time and $\gamma_{u,t}^c = \frac{ph_{u,t}^c}{N_0}$ is the communication SNR. The corresponding communication energy is [4]

$$E_{u,t}^{\text{comm}} = p\tau_c. \quad (7)$$

2.2.3 Propulsion Energy

The propulsion power consumption of a rotary wing UAV is modeled as

$$P^{\text{move}} = P_0 \left(1 + \frac{3\|v\|^2}{u_{\text{tip}}^2}\right) + P_1 \left(\sqrt{1 + \frac{\|v\|^4}{4v_0^4}} - \frac{\|v\|^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{1}{2}d_0\rho s_0\nu\|v\|^3, \quad (8)$$

where P_0 and P_1 are blade profile power and induced power in the hovering state, u_{tip} is the rotor blade tip speed, v_0 is the mean rotor induced velocity in hovering, d_0 is the fuselage drag ratio, ρ is the air density, s_0 is the rotor solidity, and ν is the rotor disk area. The energy consumption for UAV flight from $\mathbf{q}_{u,t-1}$ to $\mathbf{q}_{u,t}$ is

$$E_{u,t}^{\text{move}} = P^{\text{move}}\delta. \quad (9)$$

2.3 Target State Model, Extended Kalman Filter (EKF), and State Fusion

To track mobile ground targets, we model each target’s dynamics as a linear state-space system, where the state consists of the target’s 2-D position and velocity and evolves over time under zero-mean Gaussian process noise [4, 9]. Which captures the uncertainty inherent in real-world target motion while remaining tractable for estimation.

At each time slot, the UAV assigned to a target obtains a noisy measurement of the target’s range and azimuth, with the measurement noise variance inversely proportional to the sensing SNR. Because the mapping from the target state to range-azimuth measurements is nonlinear, we adopt an EKF to estimate the target state [1, 3, 4, 9]. Each UAV runs the EKF locally to produce a posterior state estimate and its associated covariance, which are then transmitted to the BS.

Since multiple UAVs may observe the same target with different sensing qualities, the BS fuses all local estimates into a single global estimate using inverse-covariance weighting [1]. This weighting naturally assigns greater influence to UAVs whose estimates have lower uncertainty, yielding a more reliable global estimate than any individual UAV could produce alone. The fused target estimates are subsequently used by the AAI module for target reassignment and by the MADRL module for trajectory decisions.

2.4 Tracking Accuracy Metric

To quantify tracking performance, we adopt the Posterior Cramér-Rao Lower Bound (PCRLB), which provides a theoretical lower bound on the mean-squared error of any unbiased target state estimator [1, 12]. Since the PCRLB depends on the system model and sensing geometry, it provides an estimator independent benchmark for the best tracking accuracy achievable under current UAV configuration.

The lower bound is derived from the Bayesian Fisher Information Matrix (BFIM), which is updated recursively over time and accumulates information from every UAV that successfully detects the target at each time slot [1, 12]. In practice, greater number of successful detection and better sensing geometry tighten the bound, leading to higher tracking accuracy. From this bound, we derive a scalar tracking accuracy metric that summarizes the estimation uncertainty of each target at each time slot [1]. This metric serves a dual role in our framework: it acts as a constraint in the optimization problem to guarantee a minimum tracking quality, and as a penalty term in the MADRL reward function to encourage the UAVs to learn trajectories that maintain accurate target tracking.

2.5 Problem Formulation

The objective of this work is to minimize the total energy consumption of all UAVs over the mission horizon by jointly optimizing their trajectories. The optimization is subject to several practical constraints: each UAV’s movement per time slot is limited by its maximum flight speed; every target must be observed by at least one UAV at each time slot to maintain continuous coverage; each UAV’s cumulative energy consumption must not exceed its onboard energy budget; all UAVs must remain within the designated operational area; and the tracking accuracy of every target, measured by the PCRLB-based metric, must stay below a predefined threshold to guarantee reliable tracking performance.

3 Proposed Approach

We plan to solve the formulated problem using a MADRL framework under the CTDE paradigm. MADRL is well-suited to this problem for several reasons. First, UAV trajectory decisions involve a continuous action space that is difficult to handle with classical optimization methods.

Second, multiple UAVs must learn coordinated behavior to jointly cover targets while minimizing energy, which naturally fits a multi-agent setting. Third, the environment is stochastic due to unpredictable target motion and evolving mission conditions, requiring policies that can adapt online rather than rely on static solutions. Specifically, we plan to adopt Multi-Agent PPO (MAPPO), which is well-suited to cooperative multi-agent tasks with continuous action space under the CTDE paradigm. The AAI module at the BS complements the MADRL framework by performing high-level target reassignment based on contextual reasoning, allowing the overall system to respond to dynamically changing mission requirements such as energy depletion or targets approaching critical zones.

4 Preliminary Results

As the current phase of the project focuses on system design and environment setup, no simulation results are available yet. The overall system architecture has been established, in which a BS coordinates a swarm of ISAC-enabled UAVs to track multiple ground targets. An energy model covering sensing, communication, and propulsion has been defined to drive the optimization. For target state estimation, a local EKF at each UAV combined with inverse-covariance weighted fusion at the BS has been adopted, with the PCRLB selected as the tracking accuracy metric that serves as both a constraint and a reward penalty. Finally, MAPPO under the CTDE paradigm has been chosen as the learning framework for trajectory optimization. Together, these design decisions form the foundation on which the simulation environment and MADRL training will be built in the next phase.

5 Current Progress

- System architecture and energy model finalized.
- Target state estimation pipeline (local EKF with BS-side inverse-covariance fusion) designed.
- PCRLB adopted as the tracking accuracy metric.
- MAPPO under CTDE selected as the learning framework.
- Simulation environment and MADRL implementation in progress.

6 Future Plans

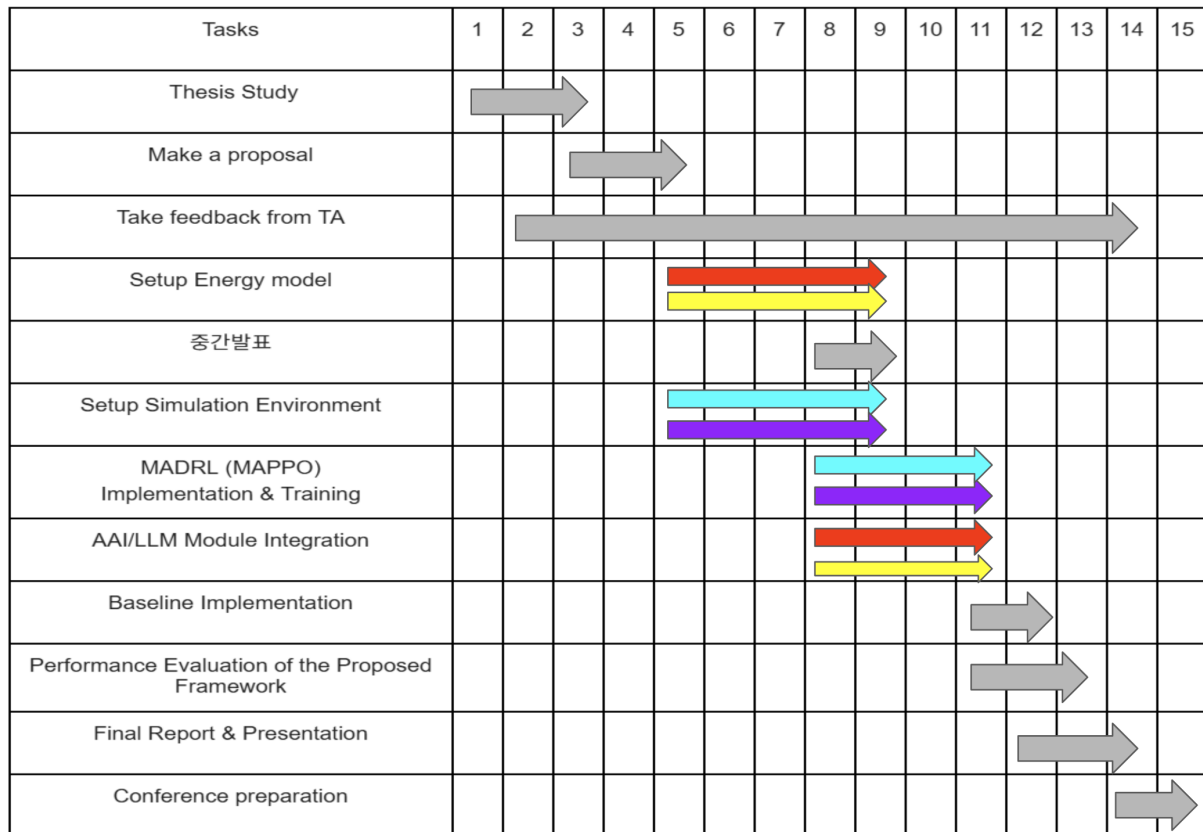


Figure 1: Project Schedule and Role Assignment

Color Legend: 공통 김지환 변혁준 신동주 이지훈

References

- [1] J. Yang, B. Wang, Z. Chang, Y. Zhao, Z. Feng, and F. Hu, "Joint trajectory planning and transmit resource optimization for multi-target tracking in multi-uav-enabled mimo radar system," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 9, pp. 11 422–11 438, 2024.
- [2] X. Zhang, N. Esfarayeni, A. B. Ayed, M. A. Chalki, P. Namaki, H. Jin, and S. Boumaiza, "Uav-based relays using active phased arrays for non-line-of-sight millimeter-wave communications: Real-time field testing," in *2024 IEEE/MTT-S International Microwave Symposium - IMS 2024*, 2024, pp. 425–428.
- [3] Z. Wei, F. Liu, C. Liu, Z. Yang, D. W. K. Ng, and R. Schober, "Integrated sensing, navigation, and communication for secure uav networks with a mobile eavesdropper," *IEEE Transactions on Wireless Communications*, vol. 23, no. 7, pp. 7060–7078, 2024.
- [4] J. Zhao, Z. Lu, and M. C. Gursoy, "Coordinated uav trajectory planning and resource allocation for target pursuit and communication," in *ICC 2025 - IEEE International Conference on Communications*, 2025, pp. 4475–4480.
- [5] L. Zhou, S. Leng, Q. Wang, and Q. Liu, "Integrated sensing and communication in uav swarms for cooperative multiple targets tracking," *IEEE Transactions on Mobile Computing*, vol. 22, no. 11, pp. 6526–6542, 2023.
- [6] T. M. Nguyen, V. T. Truong, and L. B. Le, "Agentic ai meets edge computing in autonomous uav swarms," *IEEE Internet of Things Magazine*, pp. 1–9, 2026.
- [7] W. Xie, G. Sun, R. Zhang, X. Liu, Y. Liu, J. Wang, D. Niyato, and P. Zhang, "Agentic ai for integrated sensing and communication: Analysis, framework, and case study," *arXiv preprint arXiv:2512.15044*, 2025.
- [8] Y. Liu, S. Liu, X. Liu, Z. Liu, and T. S. Durrani, "Sensing fairness-based energy efficiency optimization for uav enabled integrated sensing and communication," *IEEE Wireless Communications Letters*, vol. 12, no. 10, pp. 1702–1706, 2023.
- [9] Z. Lu and M. C. Gursoy, "Resource allocation for multi-target radar tracking via constrained deep reinforcement learning," *IEEE Transactions on Cognitive Communications and Networking*, vol. 9, no. 6, pp. 1677–1690, 2023.
- [10] Y. Zhou and X. Liu, "Trade-off between radar sensing and energy consumption in integrated sensing, computing, and communication uav network," *IEEE Transactions on Green Communications and Networking*, vol. 10, pp. 511–521, 2026.
- [11] H. Mei, H. Zhang, X. Zhou, and J. Wang, "Aoi minimization for air-ground integrated sensing and communication networks with jamming attack," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 8, pp. 12 776–12 790, 2025.
- [12] H. Zhang, W. Liu, Q. Zhang, L. Zhang, B. Liu, and H.-X. Xu, "Joint power, bandwidth, and subchannel allocation in a uav-assisted dfrc network," *IEEE Internet of Things Journal*, vol. 12, no. 9, pp. 11 633–11 651, 2025.